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Holistic Path Planning for Multi-Drone Data Collection

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***Abstract*—This paper presents how to plan for and deploy a team of Unmanned Aerial Vehicles (UAVs) to collect data from wireless sensors while minimizing collection latency. We propose a holistic solution for this problem where we first form an offline plan while considering communication and energy constraints then derive an online strategy for the UAVs to handle sensor failures and the stochastic nature of wireless communication. For offline planning, we propose a heuristic-based algorithm that turns the problem into a variation of the Vehicle Routing Problem. For our online strategy, we propose a coupled offline-online algorithm that adapts UAV behavior as needed. We evaluate our algorithms through extensive simulations as well as a proof-of-concept implementation on a physical UAV and wireless sensor testbed.**

***Index Terms*—UAV path planning, data collection**

I. Introduction

Wireless sensors are used in a variety of application areas including environmental, flora and fauna, industrial, urban, military and health monitoring. Traditionally, these sensors have been connected to form larger networks so that data can be forwarded to a central data center through multi-hop routing [1]. A drawback to this approach is that the nodes closest to data sinks become weak points in the network. To mitigate this problem, a lot of research has looked into collecting data via mobile agents, so called "data-mules" [2]. Unmanned Aerial Vehicles (UAVs) are often proposed for data-muling because they can easily traverse rugged terrain and reach areas that are challenging for humans and ground robots to access. Rotary UAVs (commonly called drones) are particularly well adapted for data collection because of their agility and the ability to hover in place while communicating with ground nodes.

Although drones have many advantages for data-muling, there are several challenges that must be considered when planning routes for data collection tasks. One major challenge in using drones is their finite on-board energy storage, requiring them to periodically recharge or change out on-board batteries. Another challenge is that drones take much longer to collect data from sensors compared with the traditional connected network approach. This motivates the need to form optimal collection paths and minimize the total data collection time.

There are several proposed path planning algorithms in recent literature for minimizing data collection time via drones. Common solutions for multi-drone scenarios include biology inspired approaches such as genetic algorithms [3] or ant colony algorithms [4], advanced Traveling Salesman Problem solvers [5], and learned methods [6]. Although some of these works consider drone energy constraints, none of them incorporate swapping out batteries to extend the range of the drones.

Another key aspect that is missing in current solutions is taking a holistic approach to solve the data-muling problem. Due to the stochastic nature of wireless communication, a data collection plan found offline may falter during deployment when sensors become weak and stop responding. Additionally, none of the existing literature has evaluated their proposed solution on a physical testbed with wireless sensors and an autonomous drone.

In this study, we formulate a holistic framework for minimizing the maximum data latency in the multi-drone data muling problem. We define *data latency* as the time lapsed from the beginning of the data collection mission until a particular piece of sensor data has been returned to the data sink.

In this paper we make the following contributions.

1) We present a holistic framework that goes from centralized, offline planning to decentralized, online control decision making for the multi-drone data muling problem.
2) We design an offline multi-drone path planning algorithm for collecting data from wireless sensors that accounts for realistic drone energy consumption and incorporates battery swaps in the planning process.
3) We propose a coupled offline-online strategy for dealing with the stochastic nature of wireless communication.
4) We evaluate our holistic approach through extensive simulations and on a physical testbed consisting of a drone and multiple wireless sensor nodes.

II. Problem Formulation

We consider a scenario with n wireless sensors deployed throughout a three-dimensional space $\mathcal{X}$ based on application needs. Let $S = \{s_1, s_2, \cdots, s_n\}$ be the set of all sensors, each of which is static. Let the position of s_i in three-dimensional space be $x_i \in \mathcal{X}$, which is known a priori. Each sensor is gathering application-specific data that must be collected by a drone and returned to a data center located at $x_b \in \mathcal{X}$. Let $A = \{a_1, a_2, \cdots, a_k\}$ be a set of k drones that will be launched from the data center at x_b and must collect all of the captured data from the sensors and return it to x_b. We assume that

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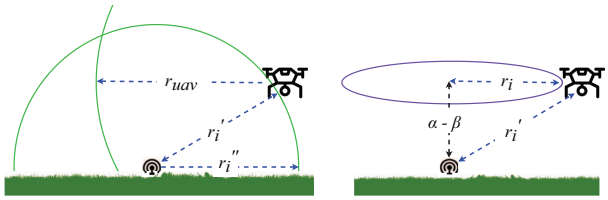

Fig. 1: Example of the service range of a sensor. The green circles represent the communication ranges for sensor i and the drone, respectively, while the purple ellipse represents the data transmission range.

each drone will maintain a constant altitude (flight-level) α. We formally define a drone sub-tour $\mathcal{T}$ as an ordered sequence of points in $\mathcal{X}$ that start and end with x_b.

A. Communication Model

We define *servicing* a sensor as a drone connecting wirelessly with a sensor and collecting all data held on that sensor. We define the *communication range* of device a as the farthest straight-line distance that another device b can be away from a and still receive data packets from a. To service a sensor, the drone and sensor must be within each other's communication range, which we term the *data transmission range*. Given communication ranges r_i'' and r_{uav} for sensor s_i and a drone, respectively, then the data transmission range is $r_i' = min(r_i'',\ r_{uav})$. Suppose that s_i is deployed at altitude β and the drone's fixed operating altitude is α. We define the *service range* as the distance away from the zenith of s_i at altitude α that the drone must be to communicate with s_i. This makes s_i's service range $r_i = \sqrt{r_i'^2 - (\alpha - \beta)^2}$. Figure 1 depicts these varying distances. We assume that r_i'' and r_{uav} are both known a priori because they can be measured experimentally while deploying the sensors.

The wireless sensors in S have varying volumes of data to be collected. Let t_i be the amount of time that a UAV must be within the service range of sensor s_i to collect all of the data from s_i. Determining t_i can be done using methods such as the *Transmission Model* described in [5]. For simplicity, we disregard how varying distances may impact t_i and leave this for future work.

Let $\mathcal{T}_i$ be a drone sub-tour. Each $\mathcal{T}_i$ will be accompanied by a sensor list, $\mathcal{S}_i$, of all sensors to be serviced on $\mathcal{T}_i$. The total time spent servicing the sensors on $\mathcal{T}_i$ will be

$$t_s(\mathcal{S}_i) = \sum_{l \in \mathcal{S}_i} t_l \tag{1}$$

We assume that only a single sensor will communicate with the drone at a time but acknowledge that in practice packet collision can be handled by the MAC layer protocol.

B. Energy Model

Drones have limited on-board energy, usually in the form of batteries, and can only operate for so long before needing to either swap-out batteries or recharge. Previous works have shown that drone power consumption is a function of drone speed [7], [8], which was verified by [9]. Let $\mathcal{P}(v)$ be the function that maps speed v to power. If given an approximation for $\mathcal{P}(v)$ and a battery's parameters, we could determine the total operating time of the drone. Let

$$s_t(v) = \frac{B_{rate} V_{bat}}{P(v)} \tag{2}$$

be the total operating time of the drone while moving at speed v where B_{rate} is the battery depletion rate (in amp-seconds) and V_{bat} is the battery voltage. We will assume that the drone's speed is fixed during the data collection mission (ether moving at a predetermined speed v_m or hovering, denoted as $v_h = 0$). Let s_m be the total amount of time that the drone can travel at v_m and s_h be the total amount of time that the drone can hover. Formally,

$$s_m = s_t(v_m) = \frac{B_{rate} V_{bat}}{P(v_m)}, \tag{3}$$

$$s_h = s_t(v_h) = \frac{B_{rate} V_{bat}}{P(v_h)}. \tag{4}$$

We base a drone's energy budget on potential-seconds of travel (ρ). Initially, a drone starts with s_m ρ based on a full battery. If the Euclidean distance from location x_i to x_j is $d_{i,j}$ then flying from x_i to x_j will consume $f(i,\ j) = \frac{d_{i,j}}{v_m}$ ρ. To convert time spent hovering into ρ we use the ratio $\frac{s_m}{s_h}$, so hovering for a duration of t seconds will consume $\frac{s_m}{s_h} t$ ρ. We denote t_i as the amount of time required to service sensor s_i (in real time) and $t_i' = \frac{s_m}{s_h} t_i$ as amount of ρ required to service s_i.

Each sub-tour $\mathcal{T}_i$ can be viewed as a sequence of edges, (i, j), that connect consecutive waypoints $x_i, x_j \in \mathcal{T}_i$. The total time to travel through each sub-tour $\mathcal{T}_i$ will be

$$t_d(\mathcal{T}_i) = \sum_{(i,j) \in \mathcal{T}_i} \frac{d_{i,j}}{v_m} \tag{5}$$

The total time to complete sub-tour $\mathcal{T}_i$, t_i^{tot}, will be the sum of the time to travel through the sub-tour and the time to service all sensors on the sub-tour. Mathematically,

$$t_i^{tot} = t_s(\mathcal{S}_i) + t_d(\mathcal{T}_i) \tag{6}$$

Similarly, the total energy, s_i^{tot} to complete $\mathcal{T}_i$ in ρ will be

$$s_i^{tot} = \sum_{(i,j) \in \mathcal{T}_i} f(i,\ j) + \sum_{l \in \mathcal{S}_i} t_l' \tag{7}$$

Each drone a_j will have a collection of sub-tours, $\mathcal{K}_j$, that contains one or more sub-tours and sensor list pairs, $\mathcal{T}_i$ and $\mathcal{S}_i$, respectively. Between consecutive sub-tours, the drone must have its battery swapped out at the data center. Let t_b be the time required to swap out batteries. Then the total mission duration of drone a_j will be

$$T_j^{tot} = \sum_{\{\mathcal{T}_i, \mathcal{S}_i\} \in \mathcal{K}_j} t_i^{tot} + t_b \tag{8}$$

Note that for the first sub-tour of each vehicle $t_b = 0$. We would also like to point out that t_b is only the time required

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to swap-out batteries and the distance required to travel back to the data center at the end of each sub-tour is included in each $\mathcal{T}_i$.

C. Problem Definition

We can now formally define the Multi-Drone Data-Muling Problem (MDDMP). Given a set of n sensors, data center location (x_b), and a team of k drones, find a set of drone routes such that

$$\min \max_j T_j^{tot},\ j \in A \tag{9}$$

where every sensor s_i is in an $\mathcal{S}_j$, and

$$s_j^{tot} \leq s_m,\ j \in A \tag{10}$$

III. Holistic Path-Planning Framework

This work presents a holistic framework for solving the Multi-Drone Data-Muling Problem (MDDMP) that first looks at offline planning then online adaptive strategies. Figure 2 shows an overview of our holistic framework for the data-muling problem.

We first find an offline solution to the MDDMP that finds sub-tours for individual drones while considering energy constraints. We then use a coupled offline-online strategy that first pre-processes the found sub-tours in the offline plan then deploys an online plan on the individual drones. During mission execution, the drones record mission statistics, such as actual communication ranges and data collection times, which we then use as feedback to update our approximations in the offline planner for the next time the drones are deployed.

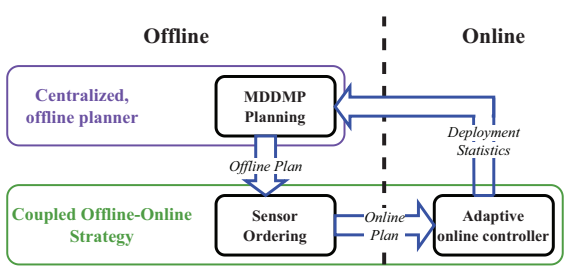

Fig. 2: Overview of our holistic framework.

A. Centralized Offline Planner

Algorithm 1 lists the details of our offline algorithm for the MDDMP. We start by fixing m, the total number of sub-tours, on line 1. We then consider the point above each sensor and treat the problem as a traditional Vehicle Routing Problem (VRP) with m vehicles, which we solve on line 3. Once we have found a solution to the VRP, we attempt to improve each sub-tour by using a variety of route tightening heuristics on lines 4-8. We then verify that our found tours are under the energy budget expressed in Equation (10) and run a job scheduling algorithm to assign sub-tours to drones on lines 9-11. We then increment the fixed number of sub-tours and repeat these steps until we either no longer see an improvement

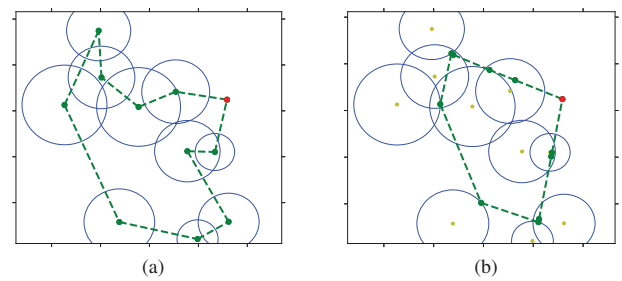

Fig. 3: Example results of the standard VRP solution (*a*) and the tightening heuristics (*b*).

in worst latency and m is a multiple of the number of drones or m has become larger than the number of sensors and return the best found set of sub-tours. We term our algorithm *VRP+T*.

There are many known approaches to solve the VRP. For this work, we propose first clustering the vertices into m clusters using Lloyd's *k*-means clustering algorithm. For each of the clusters, we add x_b (the data center's location) into the cluster and treat the cluster as an input for a Traveling Salesperson Problem (TSP) and use a heuristics-based TSP solver [10] for each problem instance.

For route tightening heuristics, we consider the service range of each sensor (the maximum distance a drone can be away from a sensor and still collect data) and attempt to move the position of each waypoint. Consider a set of three adjacent waypoints, $w_i,\ w_j,\ w_k$, on a sub-tour. We attempt to move w_j to a new position that still services s_j (the sensor that matches to w_j) using a variety of heuristics such as finding the closest point along the perimeter of the service range of s_j to the straight line between w_i and w_k. If we can find a new position for w_j that improves the total travel time, then we update w_j's position in the sub-tour and continue to cycle through the waypoints of each sub-tour until no more improvements can be found. We call these route improvement

Algorithm 1 Multi-Drone Data-Muling Algorithm

Input: S : set of sensor, x_b : data center position
Output: $\mathcal{K}$: set of drone tours

1: $m \leftarrow 1$
2: **while** *stopping condition not met* **do**
3: $\quad \mathcal{T} \leftarrow solve\text{-}VRP(S,\ x_b,\ m)$
4: $\quad$ **while** *improvement found* **do**
5: $\quad\quad$ **for** (i,j) *and* $(j,k) \in \mathcal{T}$ **do**
6: $\quad\quad\quad$ *attempt to update* j
7: $\quad\quad$ **end for**
8: $\quad$ **end while**
9: $\quad$ **if** *each* $\mathcal{T}_i \in \mathcal{T}$ *is under energy budget* **then**
10: $\quad\quad \mathcal{K}_m \leftarrow job\text{-}scheduling(\mathcal{T})$
11: $\quad$ **end if**
12: $\quad m \leftarrow m + 1$
13: **end while**
14: **return** *best found* $\mathcal{K}_m$

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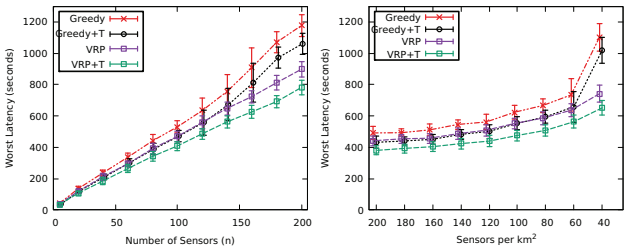
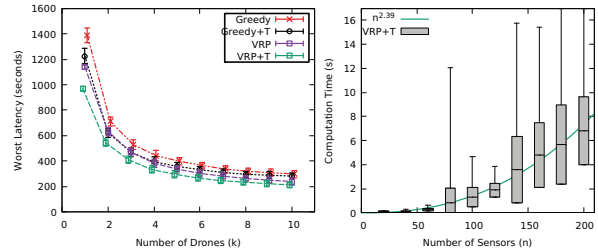


(a) (b)

Fig. 4: Graph of worst latency from simulation results while increasing the number of sensors (*a*) and the density of the sensors (*b*).



(a) (b)

Fig. 5: Graph of worst latency from simulation results while increasing the number of drones (*a*) and impact on computation time as the number of sensors increase (*b*).

steps *tightening heuristics* because they resemble pulling on a rope that runs through several fixed rings until it is tight. Figure 3 shows an example of this process.

Observe that solving the VRP will be dominated by the underlying TSP solver. The solver in [10] has an approximate time complexity of $\mathcal{O}(n^{2.2})$ and will dominate the time complexity of the tightening heuristics and common heuristics-based job-scheduling algorithms [11]. The outer while-loop will run at most n times, making the total runtime of our Multi-Drone Data-Muling Algorithm $\mathcal{O}(n^{3.2})$. However, in the average case we do not anticipate the outer while-loop to run n times, giving us a runtime between $\mathcal{O}(n^{2.2})$ and $\mathcal{O}(n^{3.2})$.

B. Coupled Offline-Online Strategy

We propose the following strategy to adapt drone behavior during deployment: Suppose that at waypoint w_i the drone must communicate with sensor s_i. If the drone is unable to communicate with s_i while at w_i, then the drone moves towards position w_i', the point directly above sensor s_i at the drone's flight-level. While moving, the drone continuously attempts to connect to s_i. If the drone connects to s_i, it stops and services the sensor. If the drone reaches w_i' without connecting to s_i, then the drone gives up attempting to service the sensor and continues the sub-tour.

When there are multiple sensors for a single waypoint that are not responding, we propose using a preassembled visit ordering. Before deploying the drone, the sensors for each waypoint can be ordered by solving a Hamiltonian path that goes from the waypoint, visits the location above each sensor, then terminates at the next waypoint on the sub-tour. We propose determining the preassembled ordering offline to avoid having the drone perform a computationally intensive task during deployment.

IV. Simulation Evaluations

In this section we evaluate our holistic solution for the Multi-Drone Data-Muling problem through simulation. We generated sets of randomized sensor deployment maps for evaluation. Unless otherwise stated, the simulation inputs have a varying number of sensors from 10 to 200 in increments of 10 (25 maps at each increment) with a random data service range from 20 *m* to 50 *m* for each sensor and an average sensor density of $\frac{150\ sensors}{1\ km^2}$.

A. Offline Algorithm Evaluation

We compare our algorithm (*VRP+T*) against three baseline approaches: (1) a greedy algorithm (*Greedy*), (2) the greedy algorithm plus our tightening heuristics (*Greedy+T*), and (3) our own algorithm without including the tightening heuristics (*VRP*).

a) **Varying Number of Sensors:** Figure 4a shows the worst data latency of running our algorithm and the baseline approaches while increasing the number of sensors with three drones. On average, our algorithm outperforms the standard *Greedy* approach by 20.7%, the *Greedy+T* approach by 10.2% and the standard *VRP* by 12.5%.

b) **Varying Sensor Density:** Figure 4b shows the worst data latency while varying sensor densities from $\frac{30\ sensors}{1\ km^2}$ up to $\frac{200\ sensors}{1\ km^2}$ with three drones and 100 sensors. Our algorithm outperforms the *Greedy* approach by 22.7%, the *Greedy+T* approach by 12.0% and the *VRP* by 13.9%.

c) **Varying Number of Drones:** Figure 5a shows the worst data latency on graphs with 100 sensors while varying the number of drones from 1 up to 10. Our algorithm outperforms the *Greedy* approach by 28.8%, the *Greedy+T* approach by 24.4% and the *VRP* by 11.0%.

d) ***VRP+T* Runtime:** Figure 5b shows the runtime of our *VRP+T* algorithm as the number of sensors increases. For reference, the function $f(n) = 2.4 \times 10^{-5} n^{2.39}$ was fitted to the runtime results. This supports our assertion that in the average case the runtime is closer to that of the TSP solver at $\mathcal{O}(n^{2.2})$ than the theoretical worst case of $\mathcal{O}(n^{3.2})$.

B. Online Strategy Evaluation

To evaluate the performance of our online strategy, we used the realistic Ardupilot Software in the Loop simulator for drone movement and the NS-3 networking simulator with transmission powers randomized over a normal distribution to simulate a stochastic network environment.

We compared our strategy against a failure agnostic approach that skips sensors when they fail to respond. Figure 6a shows the worst data latency while increasing the number

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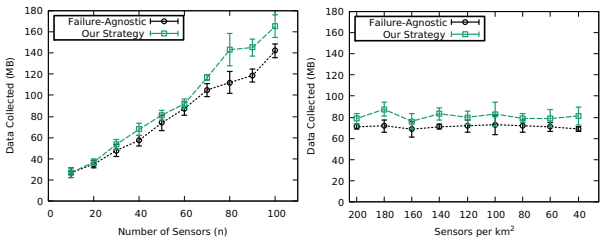

(a) (b)

Fig. 6: Graph of the volume of data collected in the realistic simulator while increasing the number of sensors (*a*) and the density of the sensors (*b*).

of sensors with a single drone. We found a 12.8% increase in total data collected when compared against the failure-agnostic approach. Figure 6b shows the worst data latency while increasing sensor density from $\frac{30\ sensors}{1\ km^2}$ to $\frac{200\ sensors}{1\ km^2}$ with 50 sensors and a single drone. When increasing sensor density we found a 14.3% increase in data collected over the baseline. This shows that our online, adaptive strategy for collecting data from wireless devices was effective in handling sensor communication failures.

V. Field Test

To validate the coupled offline-online methodology, we prototyped the problem setup on a physical testbed. For wireless sensors, we used a mix of Raspberry Pi 4's and Raspberry Pi 3B+'s. We added an additional "phantom" sensor that was not actually deployed to represent a wireless sensor that has gone dead. For the UAV, we used the platform described in [12].

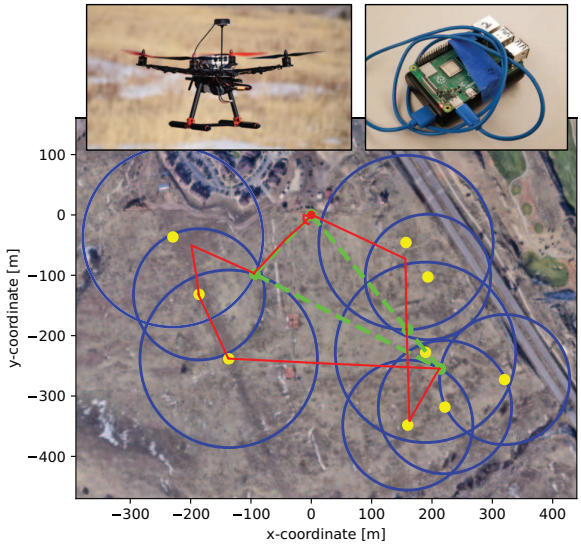

Fig. 7: Drone and one of the RPs used for prototype (top left and right, respectively) and field test results (bottom). The blue circle with yellow dots are the predicted RP service ranges and node locations. The green lines depict the offline plan while the red lines are the UAV's GPS trace during test execution.

Figure 7 shows the output of the offline plan found using our *VRP+T* algorithm and the actual GPS trace of the drone. The *VRP+T* algorithm found a solution with a worst latency of 126.3 seconds while the actual worst latency on the physical testbed was 236.3 seconds. This difference was expected because it is challenging to accurately predict communication ranges prior to deployment. The drone was able to eventually connect to all sensors except for the phantom sensor.

VI. Conclusions

In this work we presented a holistic approach to the Multi-Drone Data-Muling problem. We first solved this problem offline where we minimize data latency while applying realistic drone energy constraints and a model for limited communication range. We then proposed a coupled offline-online algorithm that adapts the drone's plan during deployment. We evaluated our framework through extensive simulations and prototyped the scenario on a physical testbed. For future work, we see a need to further improve offline planning models that more accurately compare to online deployment.

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